

Voting Management System

BSC Computer Science

**New Arts, Commerce and Science College, Ahmednagar
(Autonomous)**

In Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science (Computer Science)

Field Project Report On

**Voting Management System (VMS): A Smart Digital Voting Platform
for Secure and Transparent Elections**

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CERTIFICATE

This is to certify that the field project report entitled Voting Management System (VMS): A Smart Digital Voting Platform for Secure and Transparent Elections submitted by Mr. Aniket Jyotiram Mutake, a student of B.Sc. (Computer Science), New Arts, Commerce and Science College, Ahmednagar (Autonomous), has been completed under my supervision and guidance as part of the partial fulfillment of the requirements for the Bachelor of Science degree.

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INTRODUCTION OF ORGANIZATION

The Election Commission of India plays a vital role in ensuring free and fair elections. With the increasing voter population, it becomes essential to adopt digital solutions to enhance transparency, reduce human errors, and speed up the election process.

The Voting Management System (VMS) aims to digitalize the voting procedure, allowing voters to cast their votes securely through a web-based or local network system. The system ensures data security, transparency, and accessibility for all eligible voters.

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OBJECTIVES OF THE PROJECT

- To design a secure and transparent platform for digital voting.
- To simplify the election management process.
- To enable instant vote counting and reduce human errors.
- To enhance accessibility and ensure integrity in elections.
- To develop an environment-friendly, paperless voting system.

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PROBLEM ANALYSIS

Traditional voting systems rely heavily on manual procedures such as ballot paper printing, manual counting, and voter verification, which are time-consuming and prone to fraud. Issues like fake voting, vote duplication, and result delays often occur in such systems.

Hence, a need arises for a digital Voting Management System that eliminates these issues and provides accurate, efficient, and transparent election management.

Voting Management System

PROPOSED SOFTWARE SOLUTION: VOTING MANAGEMENT SYSTEM (VMS)

The proposed VMS is a secure web-based application that allows registered voters to log in, authenticate their identity, and cast their votes digitally. Admins can manage voter lists, add candidates, and monitor voting statistics in real time.

The system ensures high-level encryption for each vote cast and prevents multiple votes from a single user, ensuring fairness and security in elections.

Voting Management System

KEY FEATURES

1. Secure user authentication using OTP or biometric verification.
2. Online voter registration and profile management.
3. Real-time vote counting and result generation.
4. Role-based access control for admin, voters, and staff.
5. Encrypted database to protect vote confidentiality.
6. Automated report generation for election summary.
7. Mobile and desktop-friendly user interface.

Voting Management System

SYSTEM DESIGN AND IMPLEMENTATION

The VMS system follows a client-server model with a centralized database. Frontend is developed using HTML, CSS, and JavaScript, while backend functionality is powered by PHP and MySQL.

Modules:

1. Admin Module Manages users, elections, and results.
2. Voter Module Enables secure login and voting.
3. Authentication Module Validates voters through OTP or biometric verification.
4. Result Module Generates and displays results in real time.

Voting Management System

BENEFITS

- Enhances security and transparency in elections.
- Reduces human intervention and counting errors.
- Saves time and operational costs.
- Ensures real-time result declaration.
- Environment-friendly and paperless process.

CONCLUSION

The Voting Management System (VMS) represents a modern step toward digital governance and smart elections. It helps in minimizing fraud, ensuring transparency, and building public trust in the electoral process. By implementing such a system, India can move closer to efficient, fast, and corruption-free elections.